

QVM Racing Workbench Specification

Product boundary and acceptance criteria

This document defines the product boundary, domain contract, freshness policy, paper-only operating rules, validation requirements, and completion criteria for QVM Racing Workbench.

Project	QVM Racing Workbench
Document	Product specification
Status	Implementation handoff
Protected app	QVM Football Workbench
Deployment	Separate Site only after explicit approval
Trading mode	Paper positions only

Reader action: review this document before any implementation or deployment work begins.

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Purpose

Build a separate horse-racing paper-research application from the latest saved QVM Football Workbench architecture. The new product is named **QVM Racing Workbench**. It reuses QVM's operational discipline—data snapshots, deterministic gates, Manager + Worker + Auditor reports, paper ledger, calibration, decision replay, heartbeats, and timeout policy—without reusing football's domain schema or its stored data.

This specification is for the next implementation agent. It is not authorization to create or deploy a second Sites project. Implementation may happen in an isolated copy and branch; deployment requires Andrew's explicit approval after review.

Protected source application

The existing QVM Football Workbench remains the protected reference and production application:

- Existing Site: QVM Football Workbench
- Existing public URL: <https://qvm-football-workbench.emeraldpowerways.chatgpt.site>
- Current football branch at the time of writing: `timeout-fix`
- Current saved source includes `app/page.tsx`, `db/schema.ts`, `lib/qvm.ts`, `lib/qvm-timeouts.ts`, `worker_service.py`, `worker_timeout.py`, and the Drizzle migrations under `drizzle/`.

The racing implementation must not modify the football Site, its D1 database, its SQLite database, its `.env`, its Windows worker folder, or the `timeout-fix` branch. The agent must re-check the latest saved football commit before branching; the commit hash in this document is a reference point, not a permission to assume the repository has not advanced.

Product boundary

Included in the first racing release

- Upcoming and historical race records.
- Race, runner, form, non-runner, result, and market-quote normalization.
- WIN-market probabilities and fair odds for each declared runner.
- Market-implied no-vig probabilities as a baseline.
- A calibrated rating/logit challenger model with an explicit model version.
- Time-aware stale-quote and market-status gates.
- Paper positions only, with configurable stake, exposure, slippage, commission, and daily-loss controls.
- Chronological walk-forward backtesting with leakage checks.
- QVM-style Dashboard, Race Finder, Race Detail, Paper Positions, Results/Performance, Research, Agents, and Settings surfaces.
- Optional outbound Windows worker synchronization for sources that cannot run reliably inside a hosted Site.
- Hosted pre-race batch mode that can run without a Windows worker when the selected provider is reachable from the Site runtime.

Explicitly excluded from the first release

- Live bookmaker or exchange order placement.
- In-play trading or automated execution during a race.
- Place, each-way, forecast, tricast, exotic, or multi-race strategy execution.
- Copying football fixtures, teams, predictions, quotes, paper bets, portfolios, worker heartbeats, or agent history into the racing database.
- A claim that the model is profitable or a recommendation to wager real money.
- Treating The Odds API football endpoint as a horse-racing source. Racing coverage must be verified for the selected provider; a provider adapter must fail closed when the required racing fields are missing.

Domain contract

All timestamps are UTC ISO-8601 values. `scheduled_off_at` is the planned off time. `captured_at` is when QVM received a quote. `source_updated_at` is when the provider says the market or runner data changed. A quote is actionable only when the race is not closed/suspended/settled/void, the quote is within the dynamic freshness budget, and the data snapshot was available before the race.

The normalized racing entities are:

Entity	Required identity and content
Race	Provider, provider race ID, venue, country, scheduled off time, surface, discipline, distance, going, class, field size, status, source timestamps
Runner	Race, provider runner ID, horse name, number/stall, age, sex, weight, jockey, trainer, ratings, days since last run, form, equipment, declaration status
Form snapshot	Runner, historical event date, position, field size, distance/surface/going, carried weight, figure fields, and <code>available_at</code>
Market quote	Race, runner, WIN market, provider/bookmaker/exchange, captured/source timestamps, back/lay/last-traded/SP where available, overround, liquidity, commission, market status
Prediction	Race, runner, model version, model probability, fair odds, market probability, edge, uncertainty, data-quality score, feature snapshot, calculated time
Paper position	Race, runner, market, quoted/executable price, probability, edge, stake, approval, status, settlement outcome, P&L, closing price, CLV, cost fields, audit timestamps

The first model must output one probability per declared runner and probabilities must sum to 1 within 1e-6. The initial model version is `racing-logit-v1`; the no-vig market baseline is `market-no-vig-v1`.

QVM operating contract

- Deterministic code owns probabilities, freshness, data quality, stake sizing, exposure, settlement, and the kill switch.
- Manager, Worker, and Auditor agents may summarize, rank, and recommend paper action, but they cannot loosen deterministic gates or place live orders.
- `AUTO_PAPER_TRADES` remains false by default. A paper position requires a fresh actionable quote, a valid model output, a positive post-cost edge above the configured threshold, and an approval state recorded in the audit trail.
- The worker records `cycle_id`, cycle start/finish, last successful source refresh, last attempted sync, last successful sync, counts, status, and error text. The control room displays those fields separately so a failed or partially completed cycle cannot masquerade as a current successful cycle.
- `WORKER_INTERVAL_SECONDS=300` describes the base worker cadence. `REQUEST_TIMEOUT_SECONDS=300` is a request/synchronization ceiling. Neither value is a quote-freshness policy. Freshness is calculated from time-to-off and provider timestamps.

Freshness policy

The starting policy is configurable and must be stored in one module with tests:

Time to off	Maximum quote age	Behavior
More than 30 minutes	300 seconds	Actionable if the source is healthy
10–30 minutes	60 seconds	Refresh before evaluation
2–10 minutes	30 seconds	Require a recent source update
Less than 2 minutes	10 seconds	Require stream or near-real-time data; otherwise abstain
At/after off, or suspended/closed/settled/void	0 seconds	Block paper placement

If the provider cannot supply `source_updated_at`, QVM may use `captured_at` for a distant race but must lower data quality and abstain in the close-to-off windows. Non-runners or material field changes invalidate the affected prediction until the race is re-normalized and recalculated.

Validation contract

The racing model is evaluated by race date, not by randomly splitting runner rows. Every runner in one race belongs to the same train/validation/test period. Features must be sourced from snapshots with `available_at` $\leq$ `scheduled_off_at`. Results and final closing prices are available only after the event and cannot be model inputs for that race.

The minimum report includes log loss, Brier score, calibration error, top-ranked hit rate, abstention rate, market coverage, net paper ROI after configured costs, turnover, maximum drawdown, and closing-line value. Results must be compared against the no-vig market baseline and a simple favorite baseline. A positive backtest is not sufficient to enable live execution; live execution is not part of this product.

Completion criteria

The work is complete only when:

1. QVM Football Workbench builds, tests, and deploys unchanged on its existing Site path.
2. QVM Racing Workbench has a separate clean database/binding and separate secrets; no football rows or credentials are present.
3. A deterministic demo provider can populate races, runners, quotes, predictions, a paper position, and a settlement result without an external credential.
4. A stale quote, suspended market, missing runner, invalid probability vector, non-runner update, provider failure, or risk breach produces an explicit abstention/block reason and no paper position.
5. Manager + Worker control-room timestamps advance on every worker attempt and distinguish successful sync from failed sync.
6. Chronological tests prove no future result, final price, or post-off field is used in a pre-race prediction.
7. There is no live order method, live order credential, or UI control in the first release.
8. The next agent leaves a provider-coverage note, model card, test report, environment example, worker instructions, and deployment record.